



DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

DATA SCIENCE (24AD3107) A.Y. 2026 - 2027

TITLE OF THE PROJECT	Handwritten Digit/Character Recognition	
S. No.	University ID	Name
1	2420030550	Greeshma Sai
2	2420030774	Keerthi Reddy
3	2420030784	Hasini Peddi
Name of the Guide		Dr.N.Sunanda

ABSTRACT

Handwritten digit recognition is an important application of Data Science and Machine Learning because handwritten digits can differ significantly in their shape, size, orientation, and writing style. Recognizing these variations accurately is a challenging classification problem. This project focuses on developing a handwritten digit recognition system using two classical machine learning algorithms, K-Nearest Neighbors (KNN) and Support Vector Machine (SVM), and comparing their performance. The MNIST dataset is used for the project, containing a large collection of labeled grayscale images of handwritten digits from 0 to 9. The dataset is explored and preprocessed before applying the machine learning algorithms. Preprocessing includes normalization of pixel values and dividing the data into training and testing sets. Basic exploratory data analysis and visualization techniques are also used to understand the dataset, identify patterns, and examine the distribution of different digit classes.

The KNN algorithm recognizes a new handwritten digit by measuring its similarity to training samples and assigning the class that occurs most frequently among its nearest neighbors. SVM, on the other hand, attempts to find an effective decision boundary that separates the different digit classes and classifies new samples based on this boundary. Both algorithms are trained and tested using the same dataset to provide a fair comparison of their classification performance. The models are evaluated using accuracy, precision, recall, F1-score, and confusion matrix to understand their effectiveness and identify classification errors between different digit classes. The project demonstrates the practical application of fundamental Data Science and Machine Learning concepts, including data preprocessing, exploratory analysis, similarity measures, classification, train-test splitting, and model evaluation, in solving a real-world handwritten digit recognition problem.

Signature of the Guide

HOD